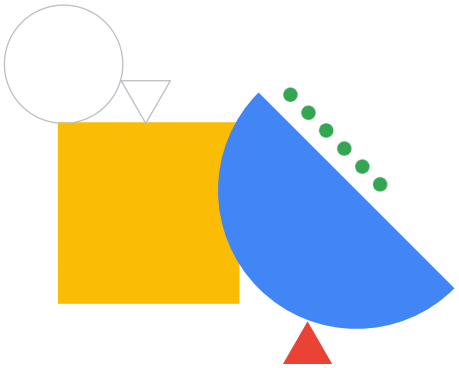
Google Cloud

Google Cloud Fundamentals: Core Infrastructure



Instructor-led training

Proprietary + Confidential



Introducing Google Cloud

Proprietary + Confidential

Introducing Google Cloud

01 An overview of cloud computing
02 IaaS and PaaS
03 The Google Cloud network
04 Environmental impact
05 Security
06 Open APIs and open source
07 Pricing and billing



Google Cloud

Let's start at the beginning with an overview of cloud computing. Some of you might be wondering about the basics: What exactly is Google Cloud? How is it organized? And what makes it unique?

In this module, we'll answer those questions.

Proprietary + Confidential

What is Cloud Computing?



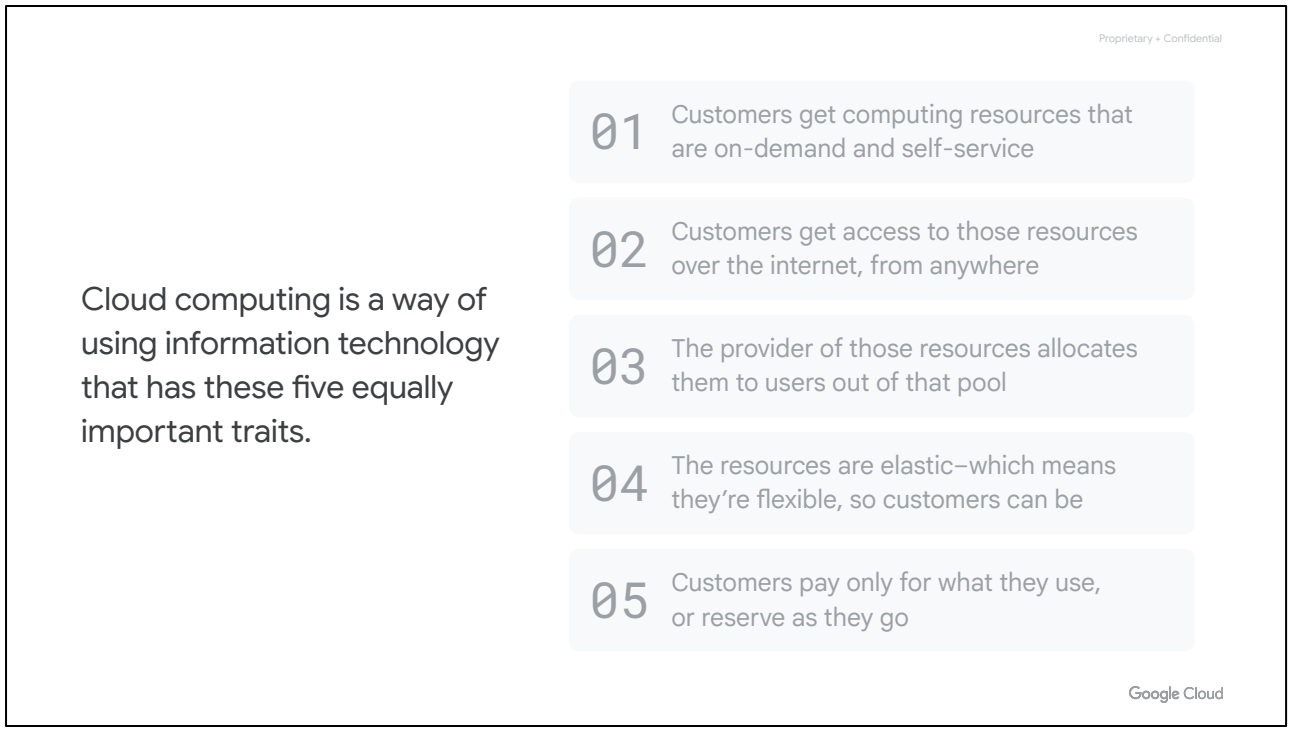
US National Institute of Standards and Technology

Cloud computing

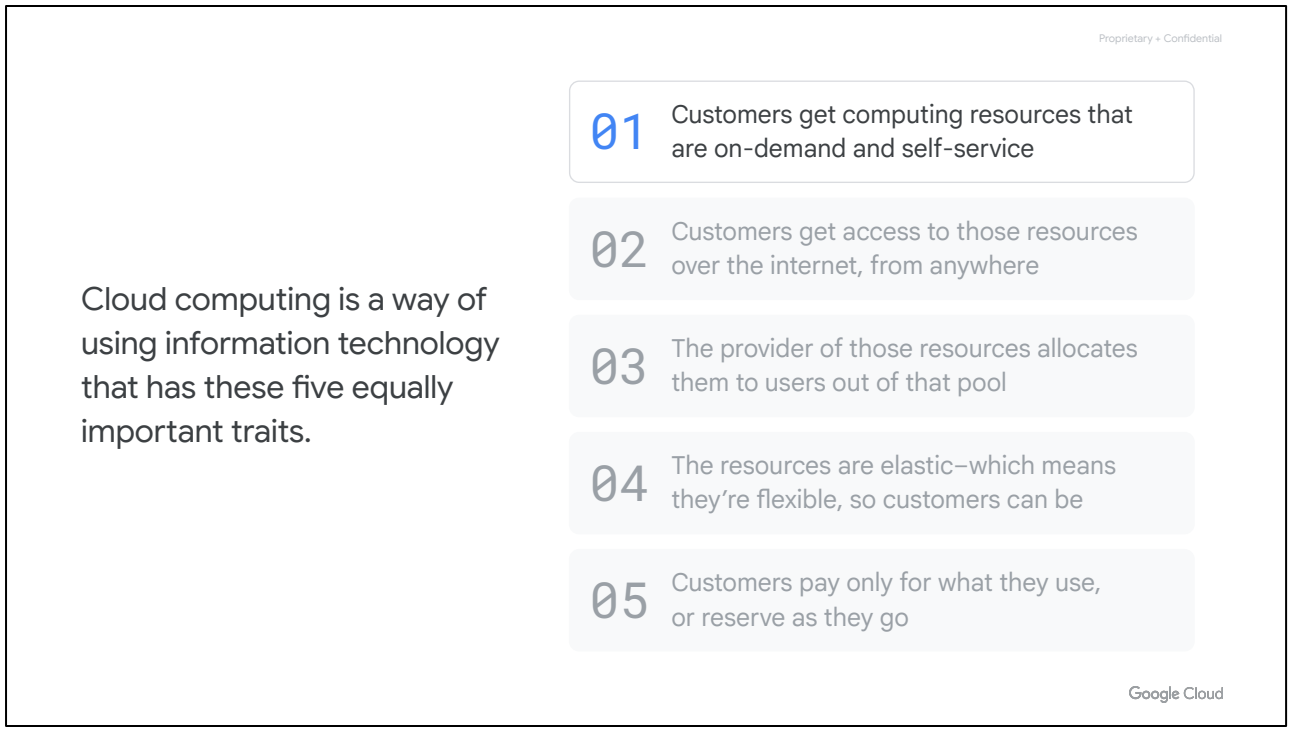
Google Cloud

The cloud is a hot topic these days, but what exactly is it?

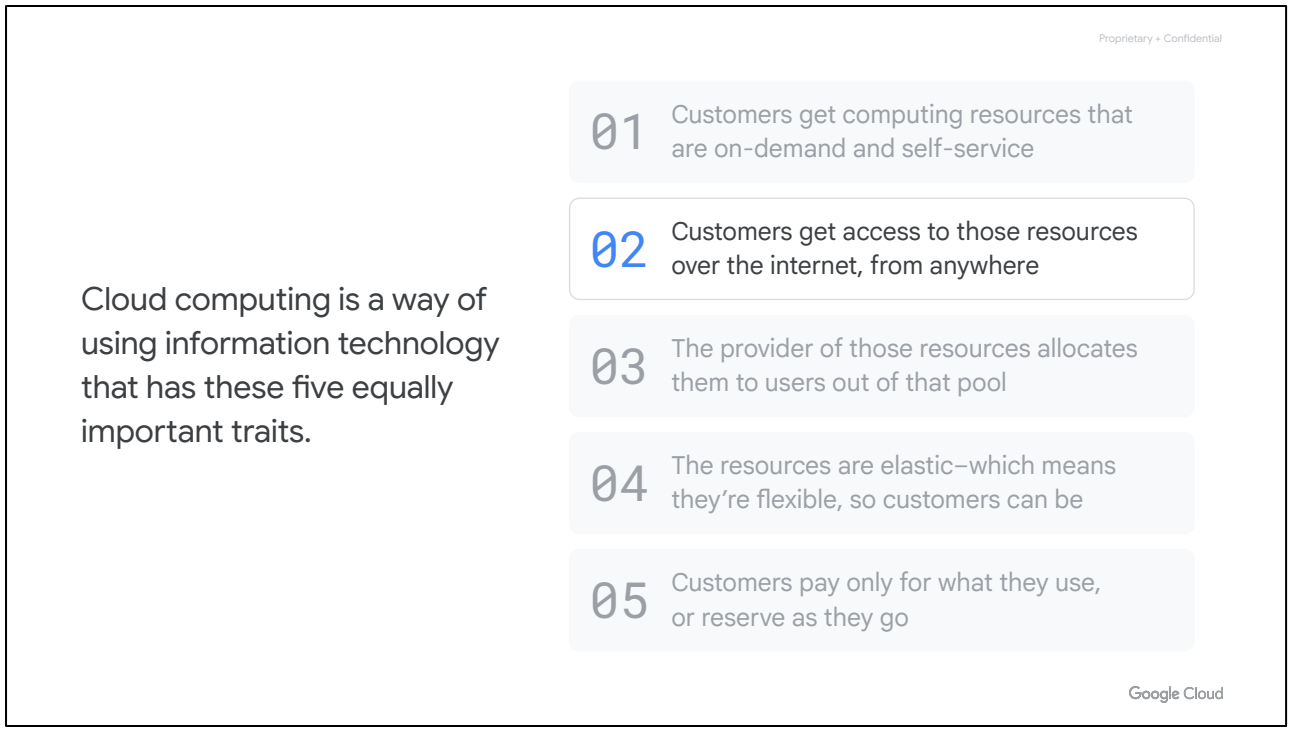
The US National Institute of Standards and Technology created the term cloud computing, although, there is nothing US-specific about it.



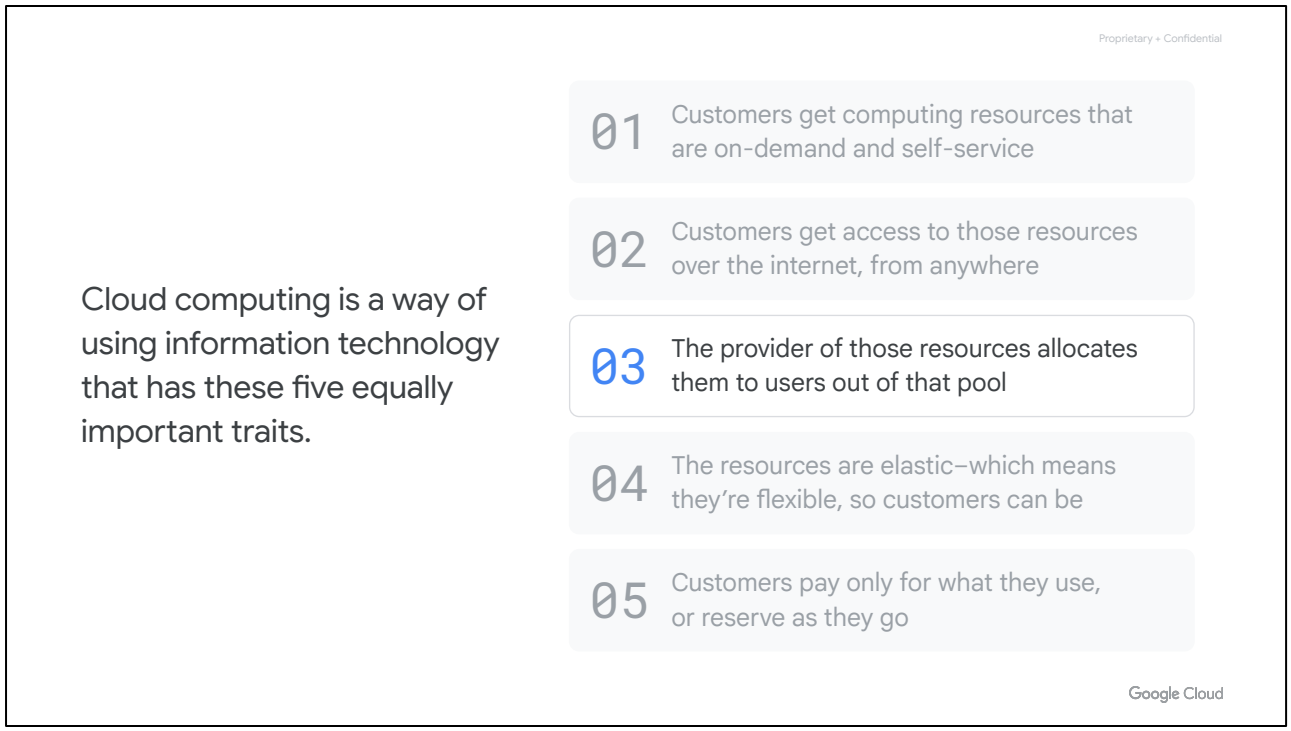
Cloud computing is a way of using information technology (IT) that has these five equally important traits.



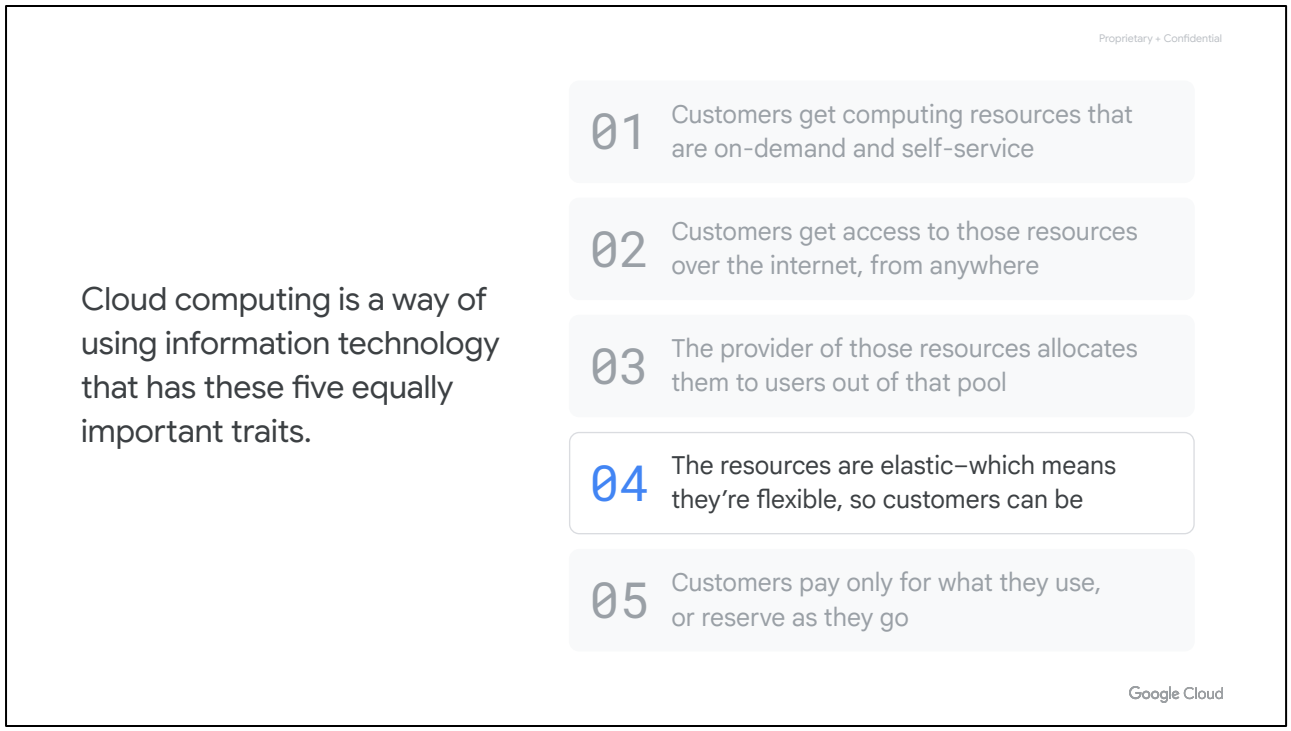
First, customers get computing resources that are on-demand and self-service. Through a web interface, users get the processing power, storage, and network they need with no need for human intervention.



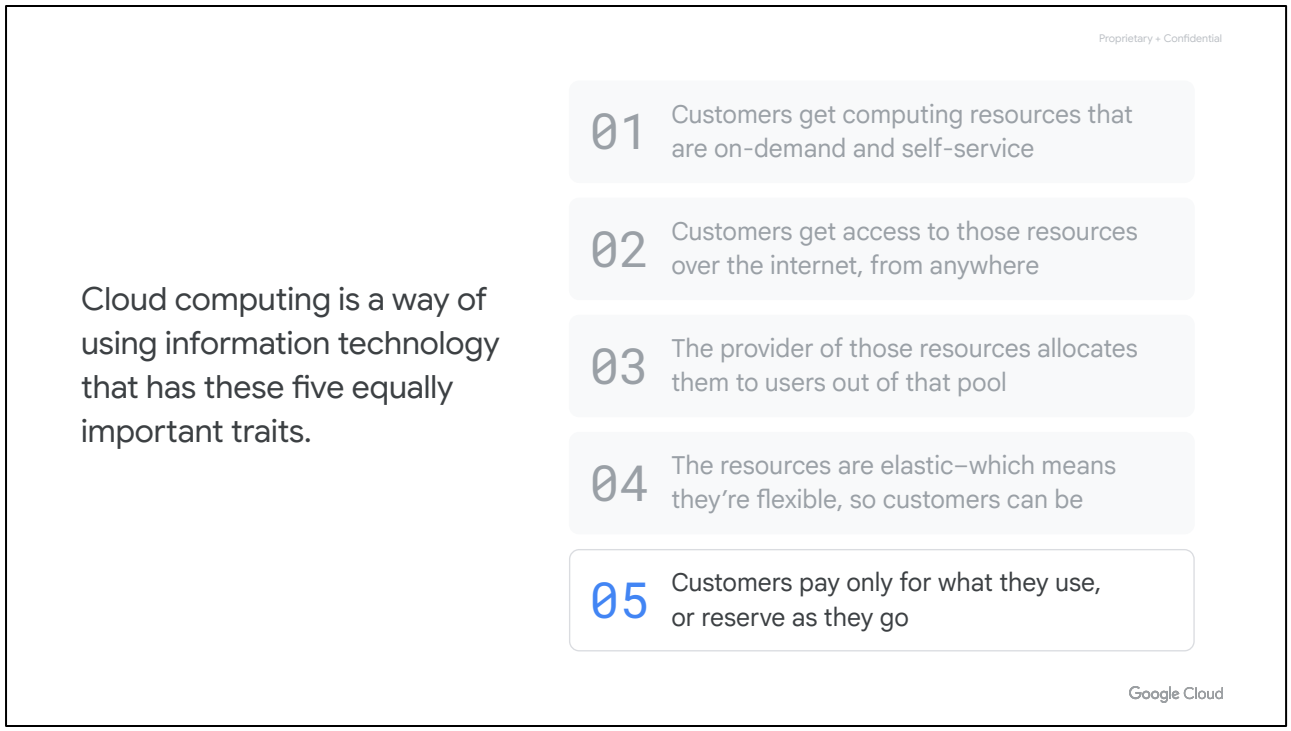
Second, customers get access to those resources over the internet, from anywhere they have a connection.



Third, the provider of those resources has a big pool of them, and allocates them to users out of that pool. That allows the provider to buy in bulk and pass the savings on to the customers. Customers don't have to know or care about the exact physical location of those resources.



Fourth, the resources are elastic–which means they're flexible, so customers can be. If they need more resources they can get more, and quickly. If they need less, they can scale back.

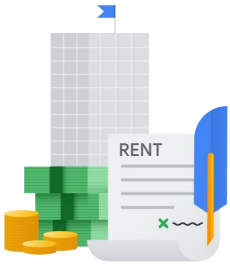


And **finally**, the customers pay only for what they use, or reserve as they go. If they stop using resources, they stop paying.

That's it. That's the definition of cloud.

Proprietary + Confidential

The history of cloud computing



First wave

Colocation

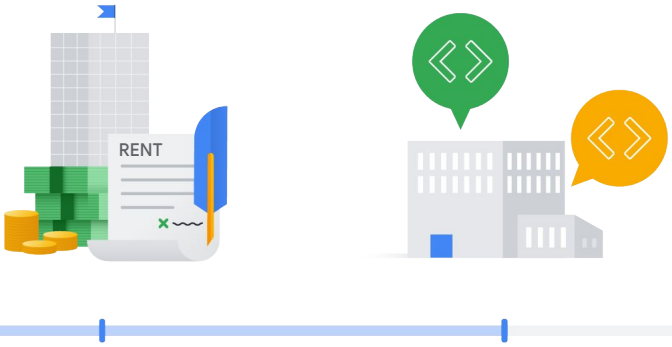
Google Cloud

But why is the cloud model so compelling nowadays? To understand why, we need to look at some history.

The trend towards cloud computing started with a **first wave** with something called *colocation*. Colocation gave users the financial efficiency of renting physical space, instead of investing in data center real estate.

Proprietary + Confidential

The history of cloud computing



First wave

Colocation

Second wave

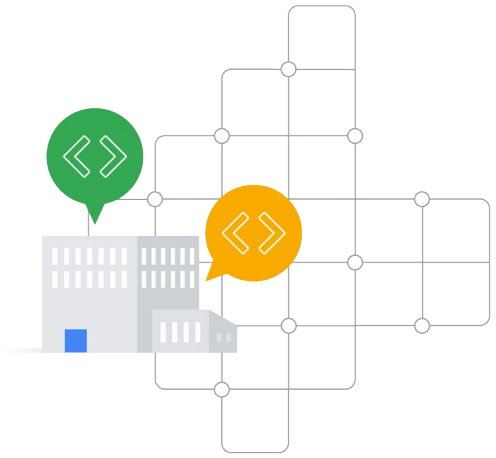
Virtualized data center

Google Cloud

Virtualized data centers of today, which is the **second wave**, share similarities with the private data centers and colocation facilities of decades past. The components of virtualized data centers match the physical building blocks of hosted computing—servers, CPUs, disks, load balancers, and so on—but now they're virtual devices.

Proprietary + Confidential

Enterprises still maintain the infrastructure



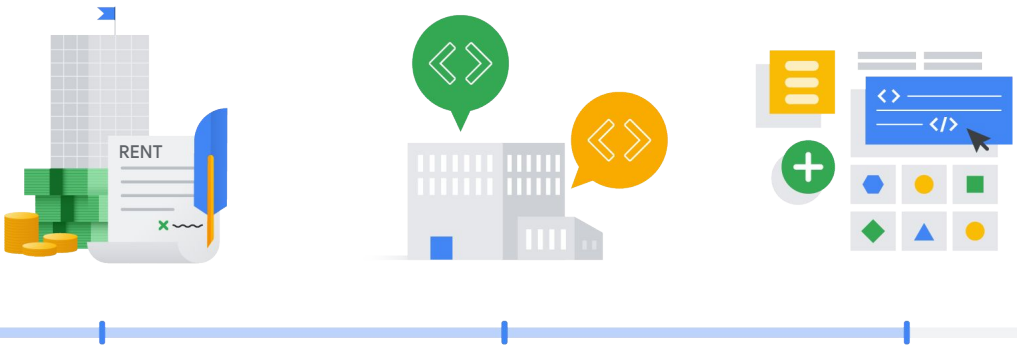
- User-controlled environment
- User-configured environment

Google Cloud

With virtualization, enterprises still maintained the infrastructure; it's still a user-controlled and user-configured environment.

Proprietary + Confidential

The history of cloud computing



First wave
Colocation

Second wave
Virtualized data center

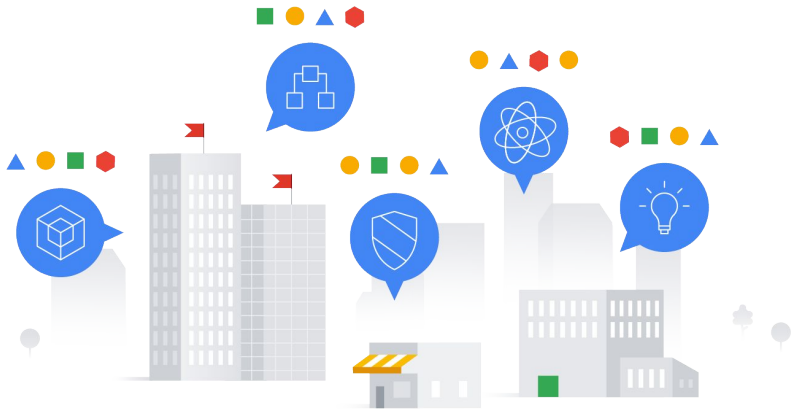
Third wave
Container-based architecture

Google Cloud

Several years ago, Google realized that its business couldn't move fast enough within the confines of the virtualization model. So Google switched to a container-based architecture—a fully automated, elastic **third-wave** cloud that consists of a combination of automated services and scalable data. Services automatically provision and configure the infrastructure used to run applications.

Proprietary + Confidential

Third-wave cloud is available to Google customers



Google Cloud

Today, Google Cloud makes this third-wave cloud available to Google customers.

Google believes that, in the future, every company—regardless of size or industry—will differentiate itself from its competitors through technology. Increasingly, that technology will be in the form of software. Great software is based on high-quality data. This means that every company is, or will eventually become, a data company.

Proprietary + Confidential

Introducing Google Cloud

01 An overview of cloud computing

02 IaaS and PaaS

03 The Google Cloud network

04 Environmental impact

05 Security

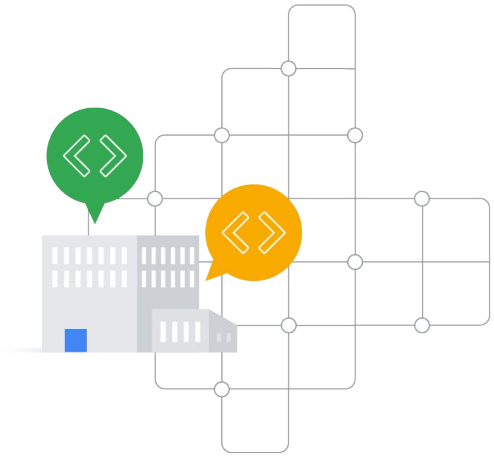
06 Open APIs and open source

07 Pricing and billing



Proprietary + Confidential

Cloud service offerings



- **IaaS** - Infrastructure as a service
- **PaaS** - Platform as a service

Google Cloud

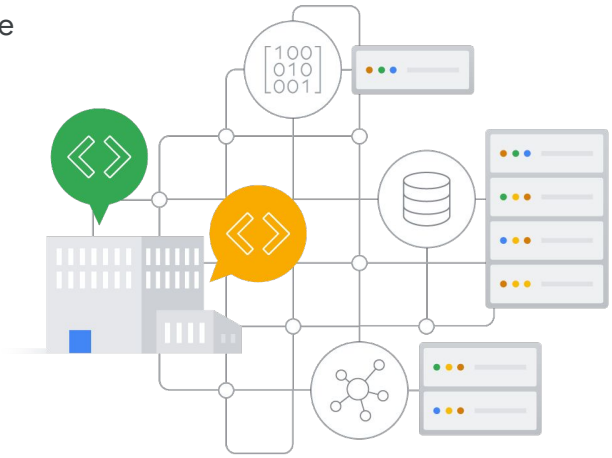
This move to virtualized data centers introduced customers to two new types of offerings: infrastructure as a service, commonly referred to as **IaaS**, and platform as a service, or **PaaS**.

Proprietary + Confidential

Infrastructure as a Service (IaaS)

IaaS delivers on-demand infrastructure resources via the cloud, such as:

- Raw compute
- Storage
- Network capabilities



Google Cloud

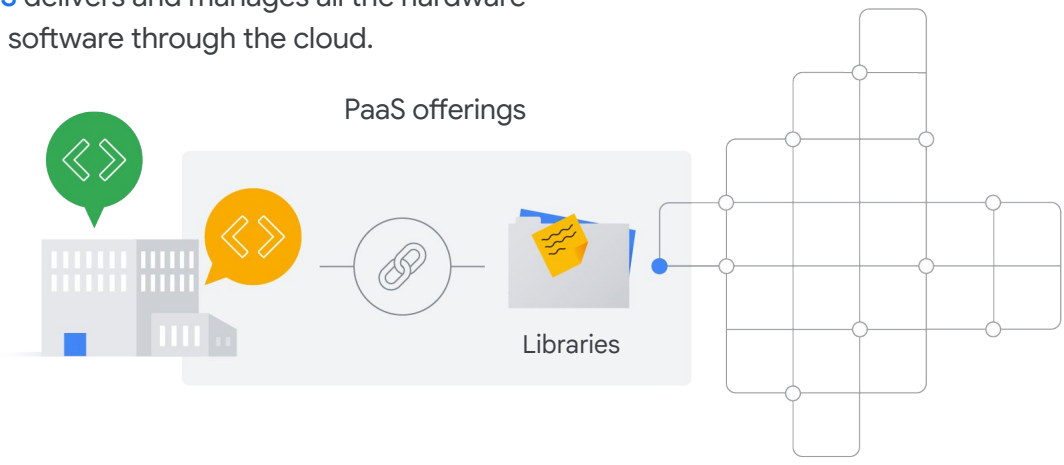
IaaS offerings provide raw compute, storage, and network capabilities, organized virtually into resources that are similar to physical data centers.

Compute Engine is an example of a Google Cloud IaaS service.

Proprietary + Confidential

Platform as a Service (PaaS)

PaaS delivers and manages all the hardware and software through the cloud.



Google Cloud

PaaS offerings, on the other hand, bind code to libraries that provide access to the infrastructure application needs. This allows more resources to be focused on application logic.

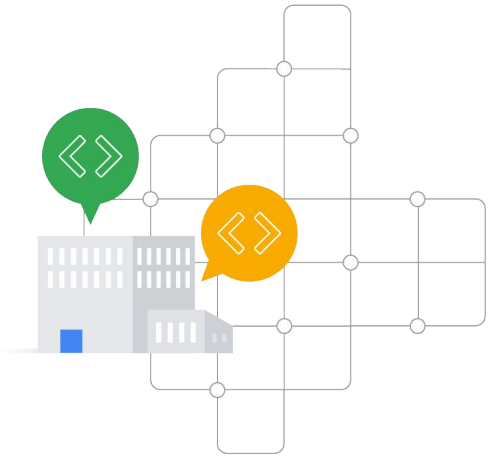
App Engine is an example of a Google Cloud PaaS service.

Proprietary + Confidential

Payment models

IaaS

Pay for what they allocate



PaaS

Pay for what they use

Google Cloud

In the IaaS model, customers pay for the resources they allocate ahead of time; in the PaaS model, customers pay for the resources they actually use.

Proprietary + Confidential

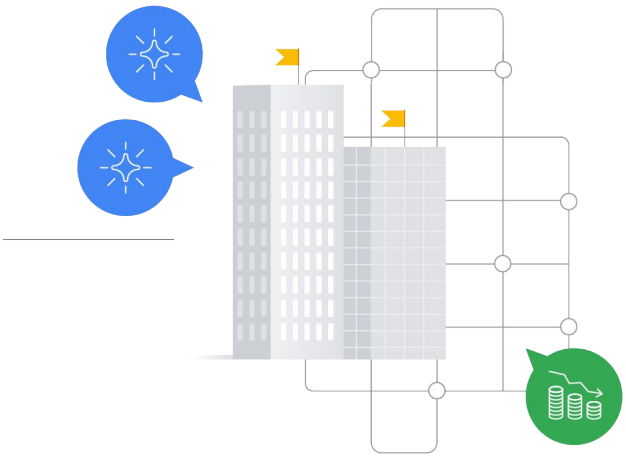
The evolution of cloud computing

Managed resources

Managed services

Deliver products/services

- More quickly
- More reliably



Google Cloud

As cloud computing has evolved, the momentum has shifted toward managed infrastructure and managed services.

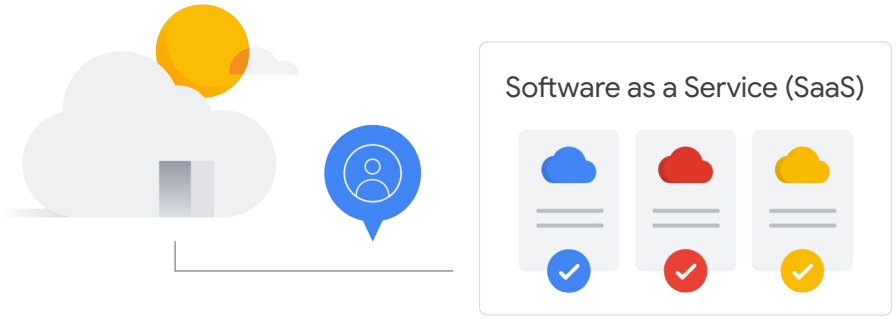
Leveraging managed resources and services allows companies to concentrate more on their business goals and spend less time and money on creating and maintaining their technical infrastructure. It allows companies to deliver products and services to their customers more quickly and reliably.

Although not shown on this slide, **serverless** is yet another step in the evolution of cloud computing. Serverless computing allows developers to concentrate on their code, rather than on server configuration, by eliminating the need for any infrastructure management. Serverless technologies offered by Google include Cloud Run, which allows customers to deploy their containerized microservices based application in a fully-managed environment, and Cloud Run functions, which manages event-driven code as a pay-as-you-go service.

Proprietary + Confidential

What about SaaS?

SaaS provides the entire application stack, delivering an entire cloud-based application that customers can access and use.



Google Cloud

You might have heard about Software as a Service (SaaS) and wondered what it is and how it fits into the Cloud ecosphere.

Software as a Service applications are not installed on your local computer - they run in the cloud as a service and are consumed directly over the internet by end users.

Google's popular applications like Gmail, Docs, and Drive, collectively known as Google Workspace, are all classified as SaaS.

You can learn more about these products or our other course offerings, but they are outside the scope of this particular course.

Proprietary + Confidential

Introducing Google Cloud

01 An overview of cloud computing
02 IaaS and PaaS
03 The Google Cloud network
04 Environmental impact
05 Security
06 Open APIs and open source
07 Pricing and billing

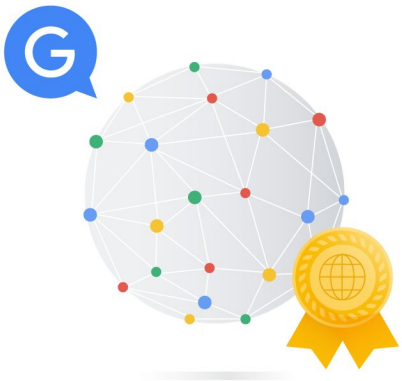


Google Cloud

Now that we've learned about or had a refresher on the basics of cloud computing, let's explore Google's own network that Google Cloud runs on.

Proprietary + Confidential

Largest network of its kind



Google's investment in its network runs into the billions of dollars

Google's network is the largest network of its kind, and Google has invested billions of dollars over the years to build it.

Proprietary + Confidential

Designed for high throughput



Google's global network is designed to give customers the highest possible throughput and lowest possible latencies for their applications by leveraging more than 100 content caching nodes worldwide–which are locations where high demand content is cached for quicker access–to respond to user requests from the location that will provide the quickest response time.

Google Cloud's locations underpin all of the important work we do for our customers. From redundant cloud regions to high-bandwidth connectivity via subsea cables, every aspect of our infrastructure is designed to deliver your services to your users, no matter where they are around the world.

Proprietary + Confidential

Infrastructure locations



Google Cloud's infrastructure is based in seven major geographic locations: North America, South America, Europe, Africa, the Middle East, Asia, and Australia. Having multiple service locations is important because choosing where to locate applications affects qualities like availability, durability, and latency, the latter of which measures the time a packet of information takes to travel from its source to its destination.

Proprietary + Confidential

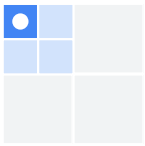
Geographic locations contain regions and zones



Location



Regions



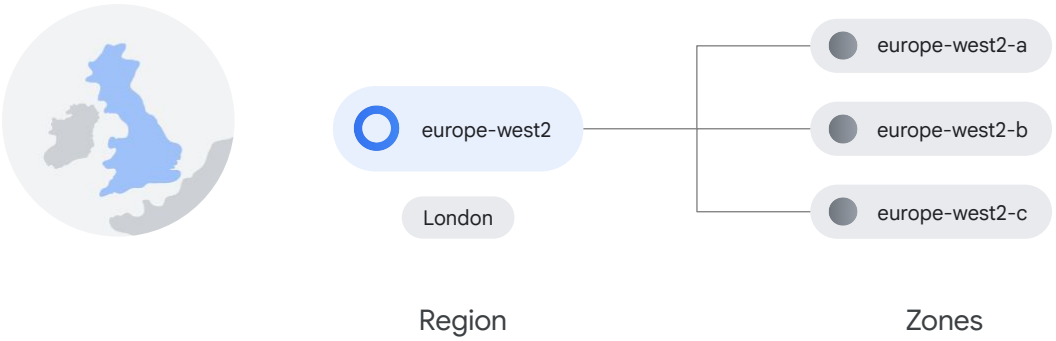
Zones

Google Cloud

Each of these locations are divided into a number of different **regions and zones**.

Proprietary + Confidential

Regions contain multiple zones

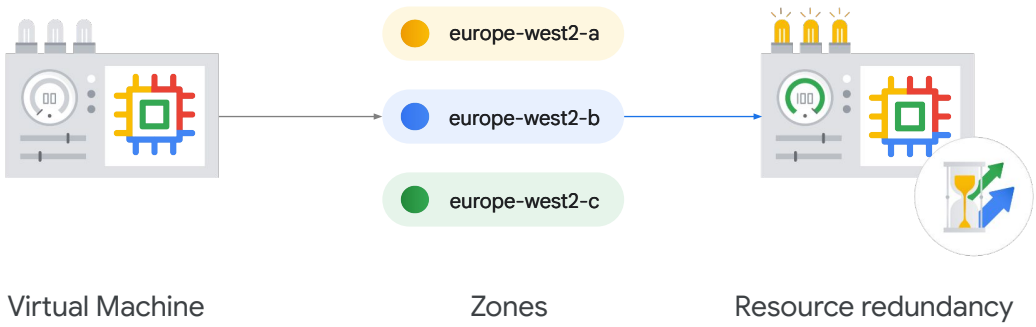


Google Cloud

Regions represent independent geographic areas, and are composed of zones. For example, London, or europe-west2, is a region that currently comprises three different zones.

Proprietary + Confidential

Zones are where Cloud resources are deployed

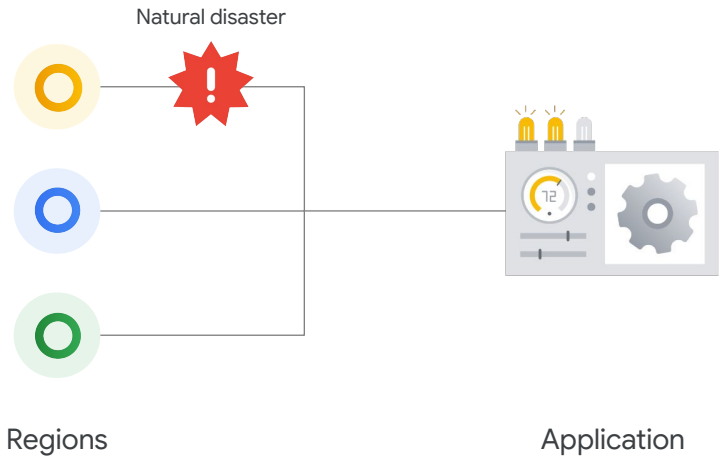


Google Cloud

A **zone** is an area where Google Cloud resources get deployed. For example, let's say you launch a virtual machine using Compute Engine–more about Compute Engine in a bit–it will run in the zone that you specify to ensure resource redundancy.

Proprietary + Confidential

Resources can run in different regions

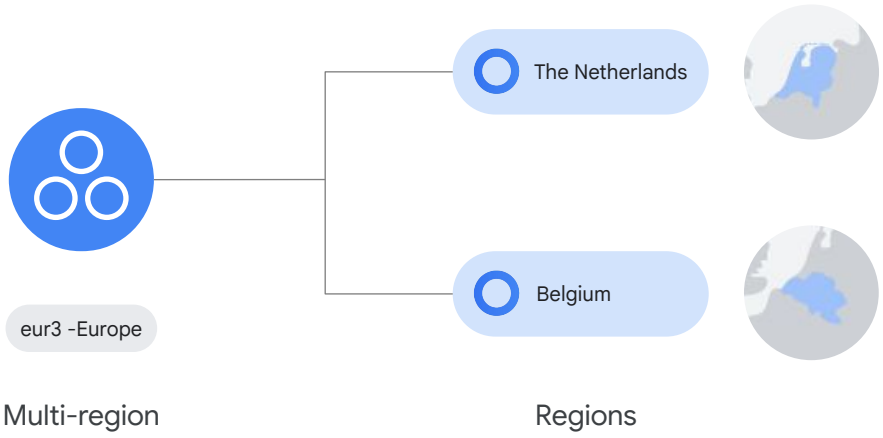


Google Cloud

You can run resources in different regions. This is useful for bringing applications closer to users around the world, and also for protection in case there are issues with an entire region, say, due to a natural disaster.

Proprietary + Confidential

Some services can run in multiple geographic locations

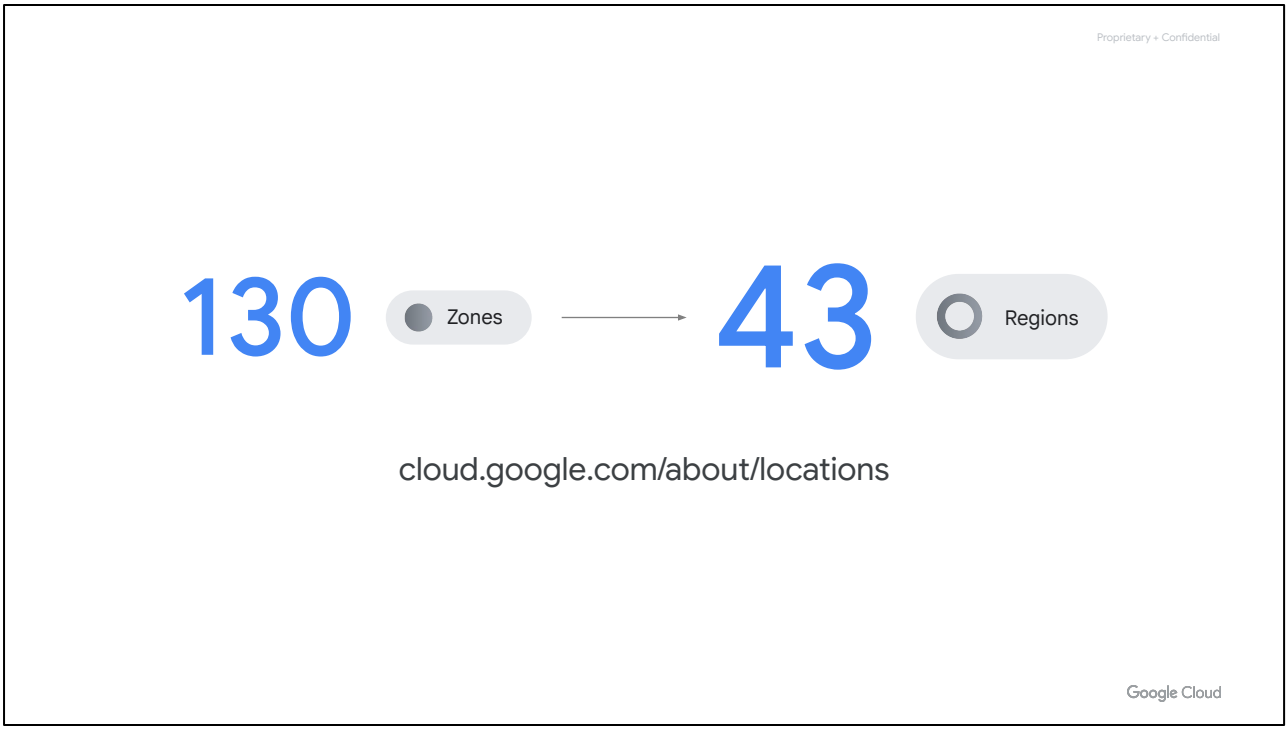


Google Cloud

Some of Google Cloud's services support placing resources in what we call a multi-region.

For example, Spanner multi-region configurations allow you to replicate the database's data not just in multiple zones, but in multiple zones across multiple regions, as defined by the instance configuration.

These additional replicas enable you to read data with low latency from multiple locations close to or within the regions in the configuration, like The Netherlands and Belgium.



The number of zones and regions Google Cloud supports is increasing all the time. The most up to date info can be found at cloud.google.com/about/locations.

Proprietary + Confidential

Introducing Google Cloud

01 An overview of cloud computing

02 IaaS and PaaS

03 The Google Cloud network

04 Environmental impact

05 Security

06 Open APIs and open source

07 Pricing and billing



Proprietary + Confidential

Data center energy consumption



2%

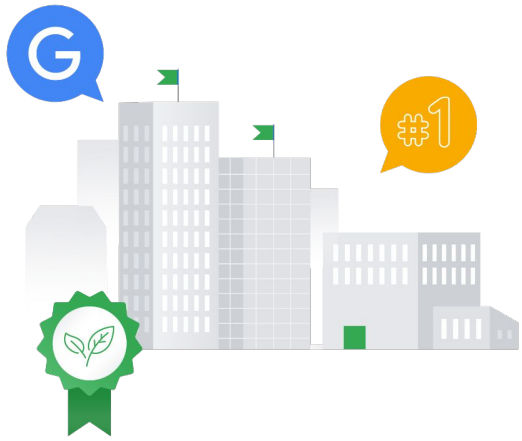
of the world's electricity

Google Cloud

The virtual world, which includes Google Cloud's network, is built on physical infrastructure, and all those racks of humming servers use huge amounts of energy. Together, all existing data centers use roughly 2% of the world's electricity. So, Google works to make data centers run as efficiently as possible.

Proprietary + Confidential

Google aims to improve efficiency and reduce waste



Google's data centers were the first to achieve ISO 14001 certification

Google Cloud

Just like our customers, Google is trying to do the right things for the planet. We understand that Google Cloud customers have environmental goals of their own, and running their workloads in Google Cloud can be a part of meeting them.

Therefore, it's important to note that Google's data centers were the first to achieve ISO 14001 certification, which is a standard that maps out a framework for improving resource efficiency and reducing waste.

Proprietary + Confidential

The data center cooling system in Finland is the first of its kind anywhere in the world.

Google's data center, Hamina, Finland



This is Google's data center in Hamina, Finland. The facility is one of the most advanced and efficient data centers in the Google fleet. Its cooling system, which uses sea water from the Bay of Finland, reduces energy use and is the first of its kind anywhere in the world.



In our founding decade, Google became the first major company to be carbon neutral. In our second decade, we were the first company to achieve 100% renewable energy. By 2030, we aim to be the first major company to operate on carbon-free energy, 24 hours a day, 7 days a week, 365 days a year.

Proprietary + Confidential

Introducing Google Cloud

01 An overview of cloud computing

02 IaaS and PaaS

03 The Google Cloud network

04 Environmental impact

05 Security

06 Open APIs and open source

07 Pricing and billing



Proprietary + Confidential

Google Cloud is designed with security in mind



Google Cloud

Because Google has nine* services with more than a billion users, you can bet security is always on the minds of Google's employees. Design for security is prevalent, throughout the infrastructure, that Google Cloud and Google services run-on.

Let's talk about a few ways Google works to keep customers' data safe.

*[Android, Chrome, Gmail, Drive, Maps, Photos, Search, Play Store, YouTube]

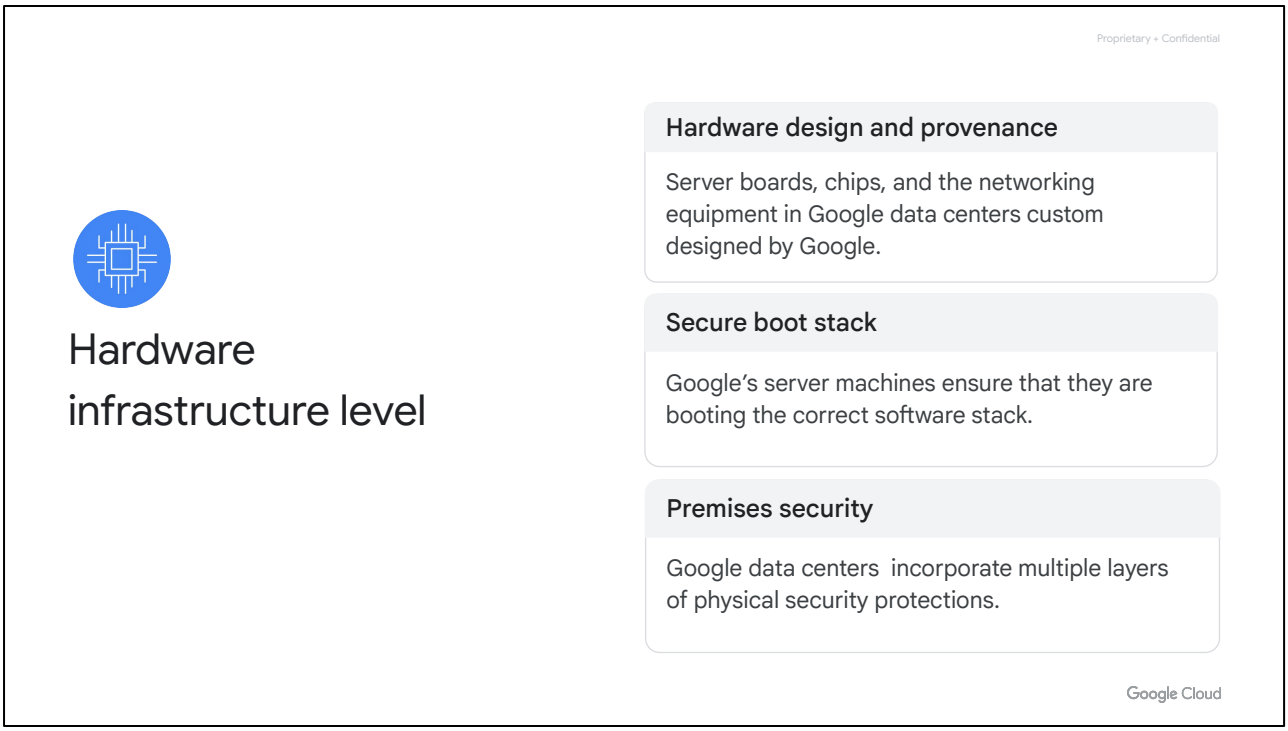
Proprietary + Confidential

Hardware infrastructure level



Google Cloud

At the **Hardware infrastructure** level:



Hardware design and provenance: Both the server boards and the networking equipment in Google data centers are custom designed by Google. Google also designs custom chips, including a hardware security chip that's currently being deployed on both servers and peripherals.

Secure boot stack: Google server machines use a variety of technologies to ensure that they are booting the correct software stack, such as cryptographic signatures over the BIOS, bootloader, kernel, and base operating system image.

Premises security: Google designs and builds its own data centers, which incorporate multiple layers of physical security protections. Access to these data centers is limited to only a very small fraction of Google employees. Google additionally hosts some servers in third-party data centers, where we ensure that there are Google-controlled physical security measures on top of the security layers provided by the data center operator.

Proprietary + Confidential

Service
Deployment level



Google Cloud

At the **Service deployment** level:

Proprietary + Confidential



Service deployment level

Encryption of inter-service communication

Services communicate with each other using remote procedure calls (“RPCs”).

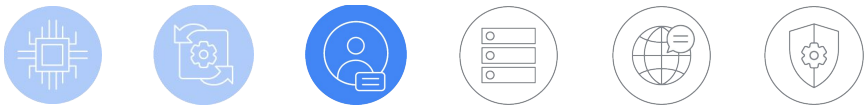
Google’s infrastructure encrypts all infrastructure RPC traffic between data centers.

Google Cloud

Encryption of inter-service communication: Google’s infrastructure provides cryptographic privacy and integrity for remote procedure call (“RPC”) data on the network. Google’s services communicate with each other using RPC calls. The infrastructure automatically encrypts all infrastructure RPC traffic which goes between data centers. Google has started to deploy hardware cryptographic accelerators that will allow it to extend this default encryption to all infrastructure RPC traffic inside Google data centers.

Proprietary + Confidential

User identity level



Google Cloud

At the **User identity** level:

Proprietary + Confidential



User
identity level

User identity

Intelligently challenges users for additional information based on certain risk factors.

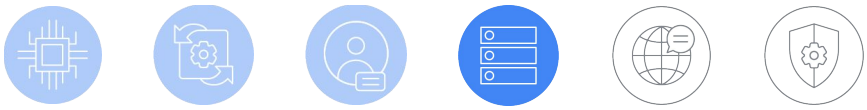
Users can use secondary factors when signing in, including Factor (U2F) open standard.

Google Cloud

User identity: Google's central identity service, which usually manifests to end users as the Google login page, goes beyond asking for a simple username and password. The service also intelligently challenges users for additional information based on risk factors such as whether they have logged in from the same device or a similar location in the past. Users also have the option of employing secondary factors when signing in, including devices based on the Universal 2nd Factor (U2F) open standard.

Proprietary + Confidential

Storage
services level



Google Cloud

At the **Storage services** level:

Proprietary + Confidential



Storage services level

Encryption at rest

Most Google applications access file storage indirectly via storage services and encryption.

Using centrally managed keys, encryption is applied at the layer of these storage services.

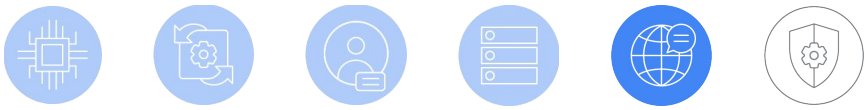
Google also enables hardware encryption support in hard drives and SSDs.

Google Cloud

Encryption at rest: Most applications at Google access physical storage (in other words, “file storage”) indirectly via storage services, and encryption (using centrally managed keys) is applied at the layer of these storage services. Google also enables hardware encryption support in hard drives and SSDs.

Proprietary + Confidential

Internet
communication level



Google Cloud

At the **Internet communication** level:

Proprietary + Confidential



Internet communication level

Google Front End ("GFE")

Google Front End ensures that all registered services use TLS connections incorporating the correct certificates and following best practices.

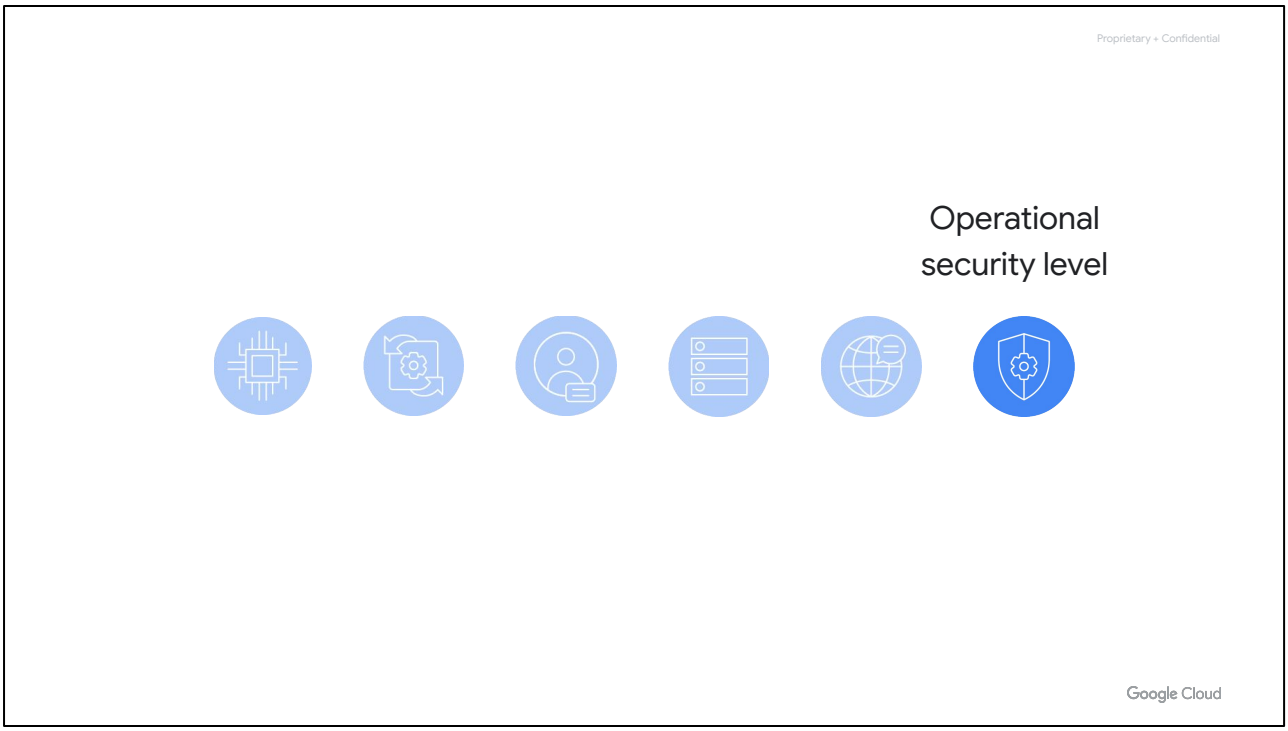
Denial of Service ("DoS") protection

Google has multi-tier, multi-layer DoS protections that reduce the risk of any DoS impact on a service running behind a GFE.

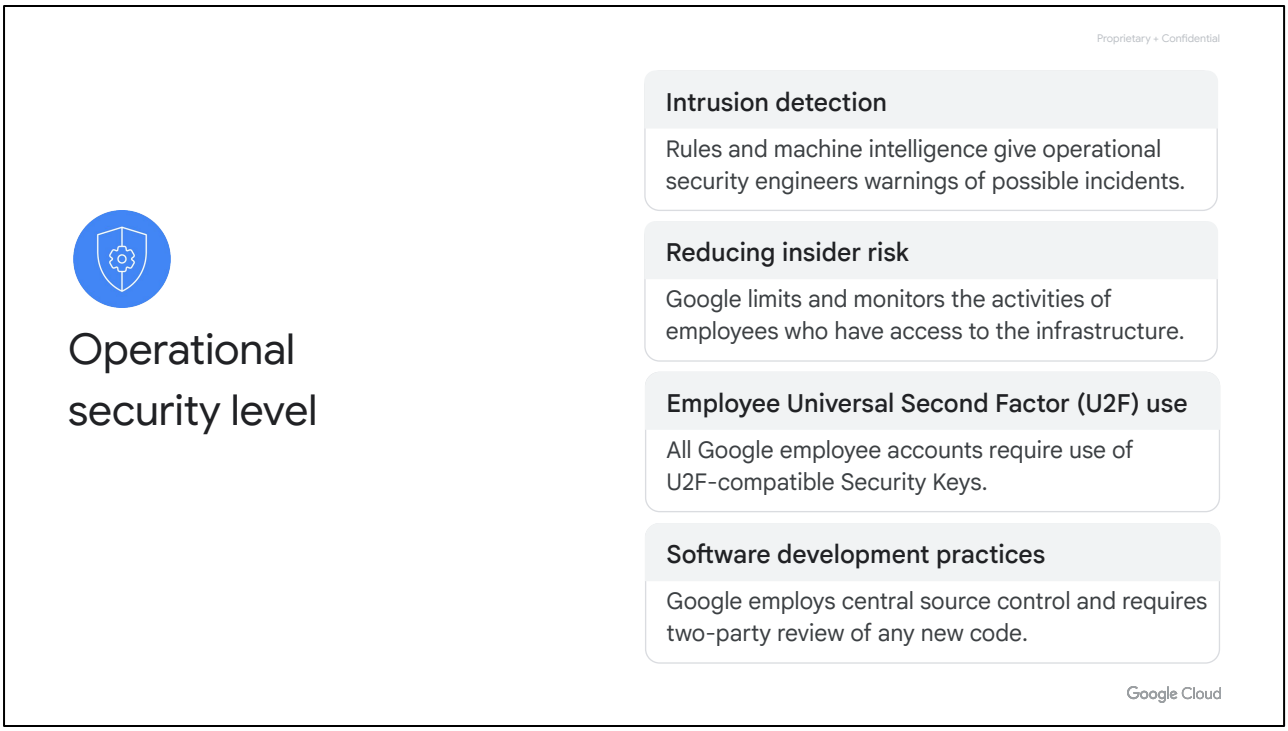
Google Cloud

Google Front End ("GFE"): Google services that want to make themselves available on the Internet register themselves with an infrastructure service called the Google Front End, which ensures that all TLS connections are ended using a public-private key pair and an X.509 certificate from a Certified Authority (CA) as well as following best practices such as supporting perfect forward secrecy. The GFE additionally applies protections against Denial of Service attacks.

Denial of Service ("DoS") protection: The sheer scale of its infrastructure enables Google to simply absorb many DoS attacks. Google also has multi-tier, multi-layer DoS protections that further reduce the risk of any DoS impact on a service running behind a GFE.



Finally, at Google's **Operational security** level:

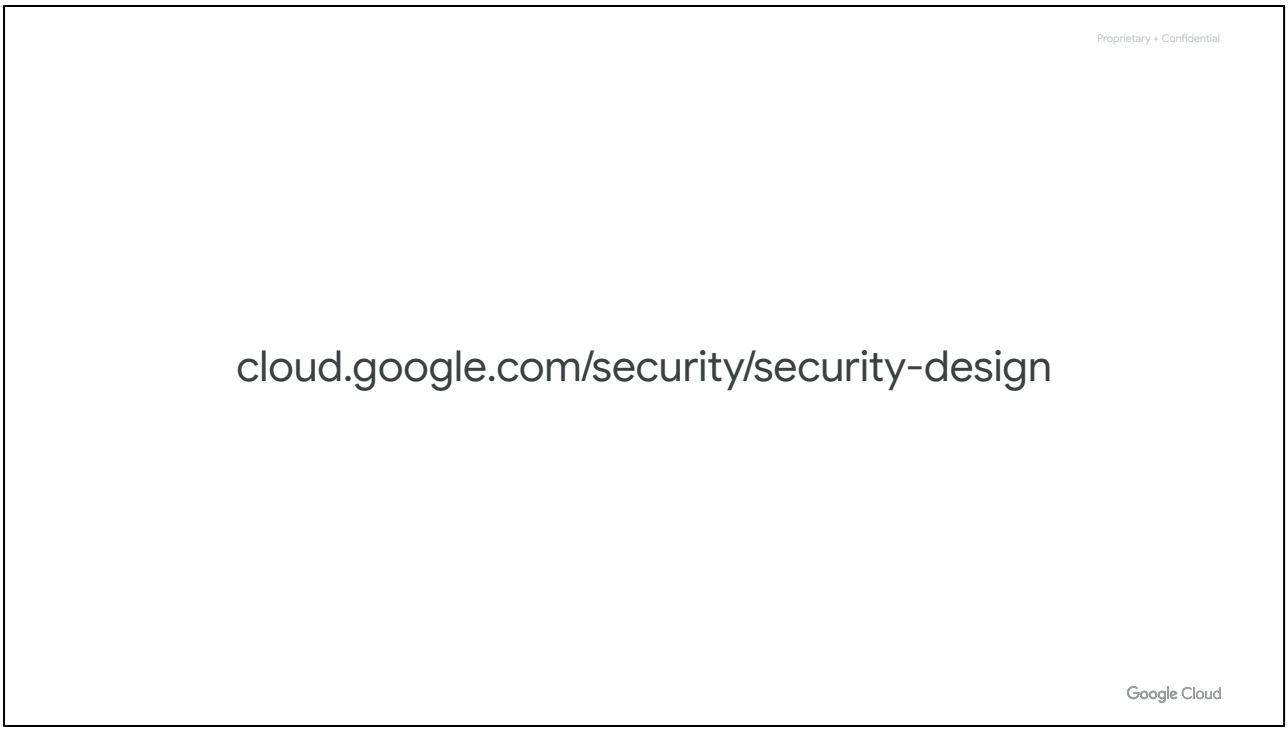


Intrusion detection: Rules and machine intelligence give Google's operational security teams warnings of possible incidents. Google conducts Red Team exercises to measure and improve the effectiveness of its detection and response mechanisms.

Reducing insider risk: Google aggressively limits and actively monitors the activities of employees who have been granted administrative access to the infrastructure.

Employee U2F use: To guard against phishing attacks against Google employees, employee accounts require use of U2F-compatible Security Keys.

Software development practices: Google employs central source control and requires two-party review of new code. Google also provides its developers libraries that prevent them from introducing certain classes of security bugs. Google also runs a Vulnerability Rewards Program where we pay anyone who is able to discover and inform us of bugs in our infrastructure or applications.



You can learn more about Google's technical-infrastructure security at cloud.google.com/security/security-design.

Proprietary + Confidential

Introducing Google Cloud

01 An overview of cloud computing
02 IaaS and PaaS
03 The Google Cloud network
04 Environmental impact
05 Security
06 Open APIs and open source
07 Pricing and billing

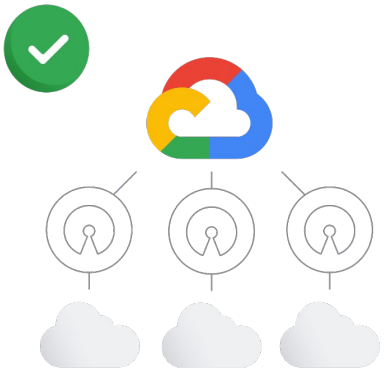


Proprietary + Confidential

Google Cloud is open-source friendly



Data locked into
a particular vendor



Google give customers the ability to run their
applications elsewhere

Google Cloud

Some people are afraid to bring their workloads to the cloud because they're afraid they'll get locked into a particular vendor.

But if, for whatever reason, a customer decides that Google is no longer the best provider for their needs, we give customers the ability to run their applications elsewhere.

Google publishes key elements of technology using open source licenses to create ecosystems that provide customers with options other than Google.

Proprietary + Confidential

Open-source TensorFlow works with Google Cloud



TensorFlow is an open source library for machine learning

TensorFlow is at the heart of a strong open source ecosystem

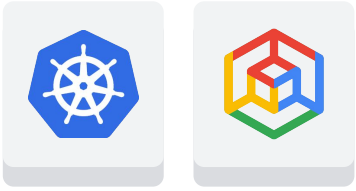
Google Cloud

For example, TensorFlow, an open source software library for machine learning developed inside Google, is at the heart of a strong open source ecosystem.

Proprietary + Confidential

Interoperability at multiple layers of the stack

Kubernetes GKE



Mix and **match microservices** running across different clouds

Google Cloud Observability



Monitor workloads across multiple cloud providers

Google Cloud

Google provides interoperability at multiple layers of the stack. Kubernetes and Google Kubernetes Engine give customers the ability to mix and match microservices running across different clouds. Google Cloud Observability lets customers monitor workloads across multiple cloud providers.

Proprietary + Confidential

Introducing Google Cloud

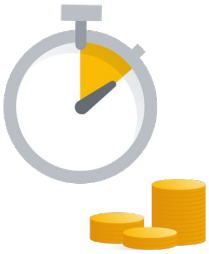
01 An overview of cloud computing
02 IaaS and PaaS
03 The Google Cloud network
04 Environmental impact
05 Security
06 Open APIs and open source
07 Pricing and billing



Now a word on Google Cloud's pricing structure.

Proprietary + Confidential

Google Compute products are billed per-second





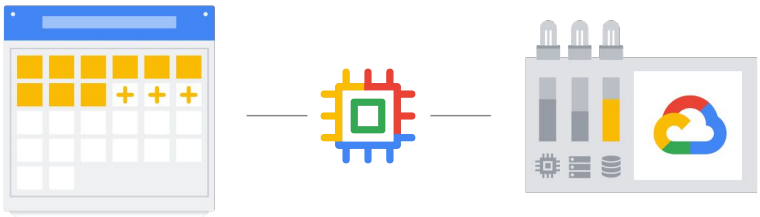
Google Cloud

Google was the first major cloud provider to deliver per-second billing for its Infrastructure-as-a-Service compute offering, Compute Engine.

We'll explore these products and services later in this course, but per-second billing is offered for users of Compute Engine, Google Kubernetes Engine (container infrastructure as a service), Managed Service for Apache Spark (which is the equivalent of the big data system Hadoop, but operating as a service), and App Engine flexible environment VMs (a platform as a service).

Proprietary + Confidential

Compute Engine offers billing flexibility



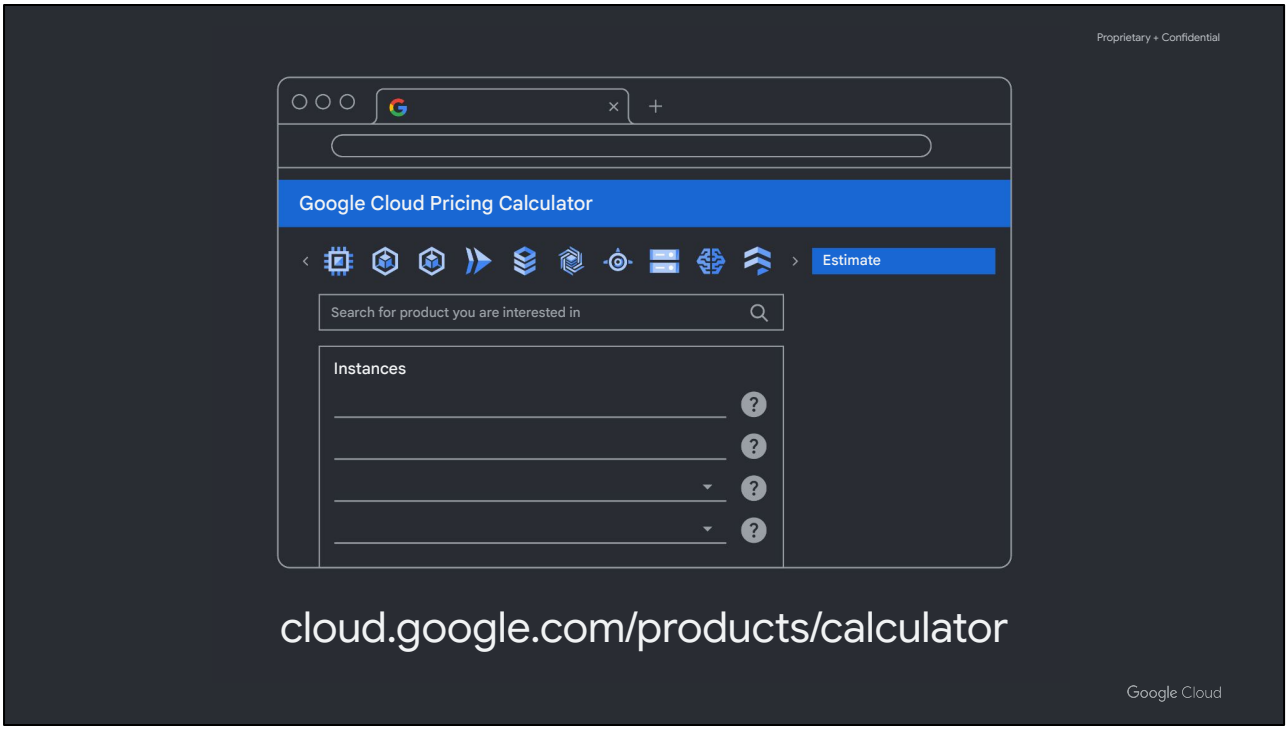
Sustained-use discounts

Custom virtual machine types

Google Cloud

Compute Engine offers automatically applied sustained-use discounts, which are automatic discounts that you get for running a virtual-machine instance for a significant portion of the billing month. Specifically, when you run an instance for more than 25% of a month, Compute Engine automatically gives you a discount for every incremental minute you use for that instance.

Custom virtual machine types allow Compute Engine virtual machines to be fine-tuned with optimal amounts of vCPU and memory for their applications, so that you can tailor your pricing for your workloads.

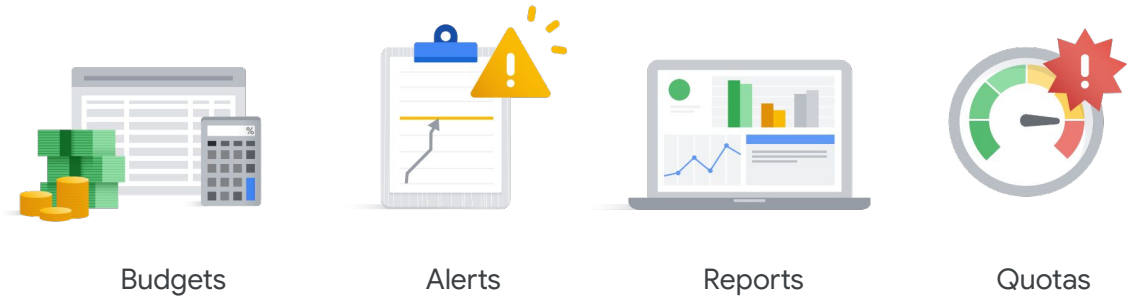


Our online pricing calculator can help estimate your costs.

Visit cloud.google.com/products/calculator to try it out.

Proprietary + Confidential

Billing tools help to budget and monitor usage



Google Cloud

You're probably thinking, "How can I make sure I don't accidentally run up a big Google Cloud bill?"

We provide a few tools to help.

1. You can define **budgets** at the billing account level or at the project level. A budget can be a fixed limit, or it can be tied to another metric - for example, a percentage of the previous month's spend.

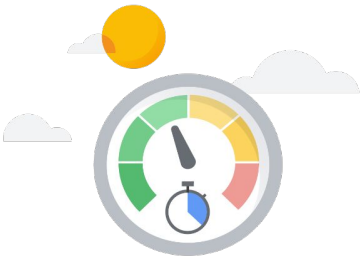
2. To be notified when costs approach your budget limit, you can create an **alert**. For example, with a budget limit of $20,000 and an alert set at 90%, you'll receive a notification alert when your expenses reach $18,000. Alerts are generally set at 50%, 90% and 100%, but can also be customized.

3. **Reports** is a visual tool in the Google Cloud console that allows you to monitor expenditure based on a project or services.

4. Finally, Google Cloud also implements **quotas**, which are designed to prevent the over-consumption of resources because of an error or a malicious attack, protecting both account owners and the Google Cloud community as a whole. It's worth noting that **quotas are actually part of Google's Identity and Access Management (IAM) tools**, but they do help with cost control.

Proprietary + Confidential

Quotas are allocated at project-level



Rate quota

Resets after
a specific time



Allocation quota

Governs the number
of resources in a project

Google Cloud

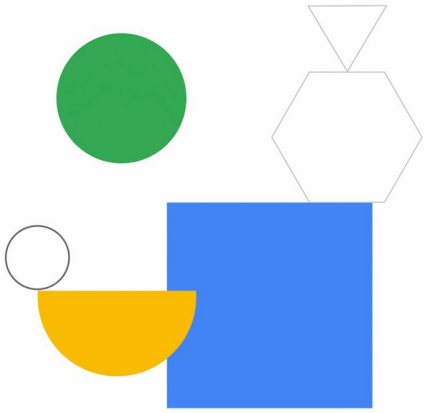
There are two types of quotas: **rate quotas** and **allocation quotas**. Both are applied at the project level.

1. Rate quotas reset after a specific time. For example, by default, the GKE service implements a quota of 3,000 calls to its API from each Google Cloud project every 100 seconds. After that 100 seconds, the limit is reset.
2. Allocation quotas govern the number of resources you can have in your projects. For example, by default, each Google Cloud project has a quota allowing it no more than 15 Virtual Private Cloud networks.

Although projects all start with the same quotas, you can change some of them by requesting an increase from Google Cloud Support.

Proprietary + Confidential

Module Quiz



Proprietary + Confidential

Quiz | Question 1

Question

What is the primary benefit to a Google Cloud customer of using resources in several zones within a region?

A. For improved fault tolerance

B. For better performance

C. For expanding services to customers in new areas

D. For getting discounts on other zones

Proprietary + Confidential

Quiz | Question 1

Answer

What is the primary benefit to a Google Cloud customer of using resources in several zones within a region?

A. For improved fault tolerance 

B. For better performance

C. For expanding services to customers in new areas

D. For getting discounts on other zones

Google Cloud

Why might a Google Cloud customer use resources in several zones within a region?

***A: For improved fault tolerance**
Feedback: Correct. As part of building a fault-tolerant application, you can spread your resources across multiple zones in a region.

B: For better performance
Feedback: That is not correct. The primary benefit of spreading your resources across multiple zones in a region is protection against failure.

C: For expanding services to customers in new areas
Feedback: That is not correct. The primary benefit of spreading your resources across multiple zones in a region is protection against failure.

D: For getting discounts on other zones
Feedback: That is not correct. The primary benefit of spreading your resources across multiple zones in a region is protection against failure.

Proprietary + Confidential

Quiz | Question 2

Question

What type of cloud computing service lets you bind your application code to libraries that give access to the infrastructure your application needs?

A. Hybrid Cloud

B. Infrastructure as a Service

C. Software as a Service

D. Platform as a Service

E. Virtualized data centers

Proprietary + Confidential

Quiz | Question 2

Answer

What type of cloud computing service lets you bind your application code to libraries that give access to the infrastructure your application needs?

A. Hybrid Cloud

B. Infrastructure as a Service

C. Software as a Service

D. Platform as a Service 

E. Virtualized data centers

Google Cloud

What type of cloud computing service lets you bind your application code to libraries that give access to the infrastructure your application needs?

A: Hybrid cloud
Feedback: Incorrect. This refers to a cloud deployment model (combining private and public clouds) rather than a service category. It describes where the cloud is, not how you interact with the code.

B: Infrastructure as a Service
Feedback: Incorrect. IaaS provides raw resources like VMs and storage. In IaaS, you are responsible for managing the OS and middleware yourself, rather than binding code to provided libraries.

C: Software as a Service
Feedback: Incorrect. SaaS provides a finished product managed by the vendor. You generally consume the application via a web browser rather than writing or binding your own application code to it.

D: Platform as a Service
Feedback: Correct! PaaS is designed for developers. It provides the SDKs, libraries, and runtime environments needed to build and deploy applications without managing the underlying hardware or OS.

E: Virtualized data centers

Feedback: Incorrect. While virtualization is a technology that powers cloud computing, it is not a cloud service model. This describes a physical infrastructure setup, often on-premises.

Proprietary + Confidential

Quiz | Question 3

Question

Why might a Google Cloud customer use resources in several regions around the world?

A. To improve security

B. To earn discounts

C. To offer localized application versions in different regions

D. To bring their applications closer to users around the world, and for improved fault tolerance

Proprietary + Confidential

Quiz | Question 3

Answer

Why might a Google Cloud customer use resources in several regions around the world?

A. To improve security

B. To earn discounts

C. To offer localized application versions in different regions

D. To bring their applications closer to users around the world, and for improved fault tolerance



Google Cloud

Why might a Google Cloud customer use resources in several regions around the world?

A: To improve security
Feedback: Running an application in several regions does not necessarily improve security. Instead, using resources in multiple regions may give better performance for global users, and it also can protect against failures of entire regions.

B: To earn discounts
Feedback: Running an application in several regions does not necessarily earn discounts. Instead, using resources in multiple regions may give better performance for global users, and it also can protect against failures of entire regions.

C: To offer localized application versions in different regions.
Feedback: Google Cloud allows you to develop different localized versions of your application even if your application is based in a single region. Running an application in several regions has no impact on the localized versions of your application.

***D: To bring their applications closer to users around the world, and for improved fault tolerance**
Feedback: That is correct.